Personalized 'You may also like' on PDP

Replacing the static 'frequently bought together' module with personalized recommendations on the product detail page increases add-to-cart rate and downstream conversion.

AT A GLANCE

Primary metric is *conversion_rate*; decision is binary against Ship if add_to_cart_rate lift exceeds MDE/2 (≥1% relative) with CI clearing zero.

Pre-registered before any analysis. Decision is binary against conversion_rate: a 20.0% relative relative lift over a 10.0% baseline requires 12,000 per arm at the standard a / power settings. Guardrails sit on revenue_per_user.

SAMPLE REQUIRED (PER ARM)

12,000

to detect a 20.0% relative relative effect at a = 0.05, power = 0.80

BASELINE
10.0%

DURATION
14 days

TOTAL SAMPLE
24,000

FEASIBILITY
VIABLE

MECHANISM

Why we think this works

Personalization based on past purchase history should surface more relevant complementary products than the static co-purchase baseline, especially for repeat customers.

The static module ships for everyone today; click-through on the module is ~6%. Personalization in the cart page (a separate test in 2025Q3) showed a 1.5pp lift in CTR.

METRICS

conversion_rate is the verdict-driving metric.

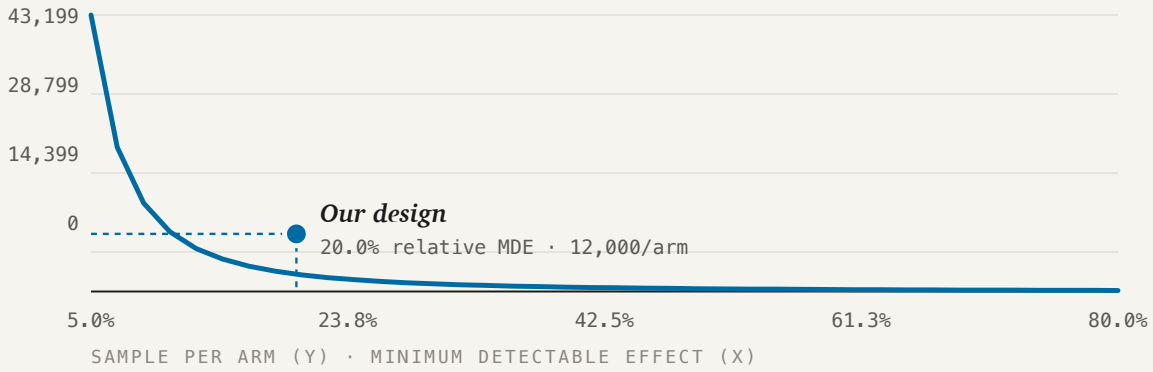
Primary is conversion_rate; secondaries inform interpretation but do not influence the verdict. Guardrails block ship if their 90% CI lies past the pre-registered harm boundary.

PRIMARY	**conversion_rate**	Proportion · baseline 10.0% · MDE 20.0% relative · direction higher is better
SECONDARY	**conversion_rate**	Leading indicator. Not used for the ship/no-ship decision but informs interpretation.
SECONDARY	**average_order_value**	Leading indicator. Not used for the ship/no-ship decision but informs interpretation.
GUARDRAIL	**revenue_per_user**	Direction higher is better. Decision blocks if 90% CI lies past -5.0% relative.

POWER

Detecting smaller effects requires much more sample.

To resolve a 20.0% relative relative effect against a 10.0% baseline, we need 12,000 per arm. Tighter detection thresholds would climb the curve to the left — quickly.



Smaller effects require disproportionately more sample. Our design point is feasible inside the available user_id surface.

ASSIGNMENT + COHORTS

First-exposure attribution at the user_id level.

Each user is randomly assigned at first exposure to control or treatment in a 50 / 50 split. Events before first_exposure_at are not attributed to the experiment.

UNIT	**user_id** One row per (user_id, experiment_id) in the assignments table; first_exposure_at gates which events are attributed.
ARMS	**50 / 50** control · treatment
COHORT	**all assigned users**

DECISION RULES

Pre-registered. The analyze verb walks the same tree against the same metrics.

Ship if add_to_cart_rate lift exceeds MDE/2 (≥1% relative) with CI clearing zero. Smaller lifts surface a caveat.

SHIP	Primary 95% CI excludes zero on benefit side · no guardrail breach · novelty late-ratio ≥ 0.7
LIFT WITH CAVEAT	Real lift but below MDE/2 — surface the caveat in the readout but still consider rolling out
NO-SHIP · GUARDRAIL	Any guardrail's 90% CI lies past its harm threshold
NO-LIFT	Well-powered, primary CI straddles zero with half-width > 2 × MDE
INCONCLUSIVE	Underpowered (n < required) AND primary CI straddles zero
INVALID	SRM χ^2 p < 0.0005 and no override resolved — investigate assignment

INTEGRITY LOCK · THREE-PART SEAL

brief.sealed.yaml binds intent + metric definitions + expected shape.

Tampering with any of the three components after seal will fail `agentxp.schemas.brief_seal.verify_or_raise`; `agentxp analyze` refuses to open a brief whose seal does not verify (R11). This is the architectural wall, not a guideline.

DESIGN_CHAIN_HASH	31c7896bb5954e42a36a3d4dcff51468e0106812c89b561e84cdadd9abcb607e
METRIC_SNAPSHOT	1 metric hashed
CONVERSION_RATE	fbf7a228872bb052817eb550deb3311acba51c3cb60f03241c460211b307f2d0
EXPECTED_SHAPE	user_id · 50 / 50
SEALED_AT	2026-06-05T15:42:19Z

SEALED_BY	walk_experiment_demo
